



Tecnológico de Monterrey

Escuela de Ingeniería y Ciencias

Monterrey Campus

Autonomy of Unmanned Aerial Vehicles

MR3003C.601

Report:

Practice 07 – Motor Velocity Control

Part B. I+P Control Strategy

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Due date:

September 3rd, 2024

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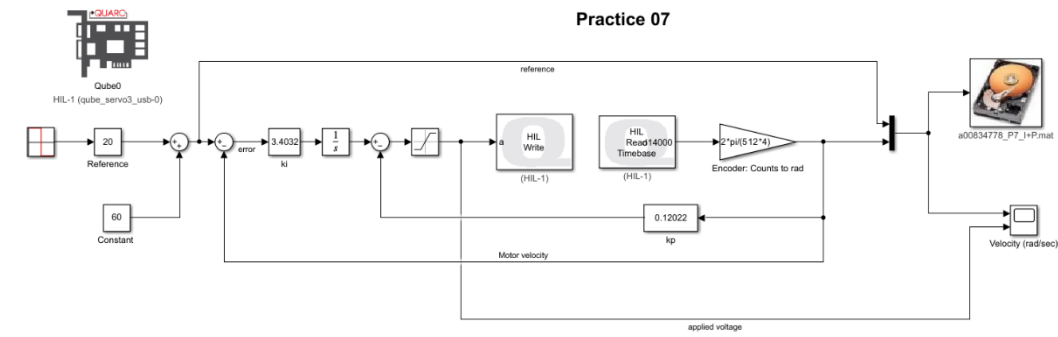
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1. Mathematical Expression for the Control Law $u(t)$

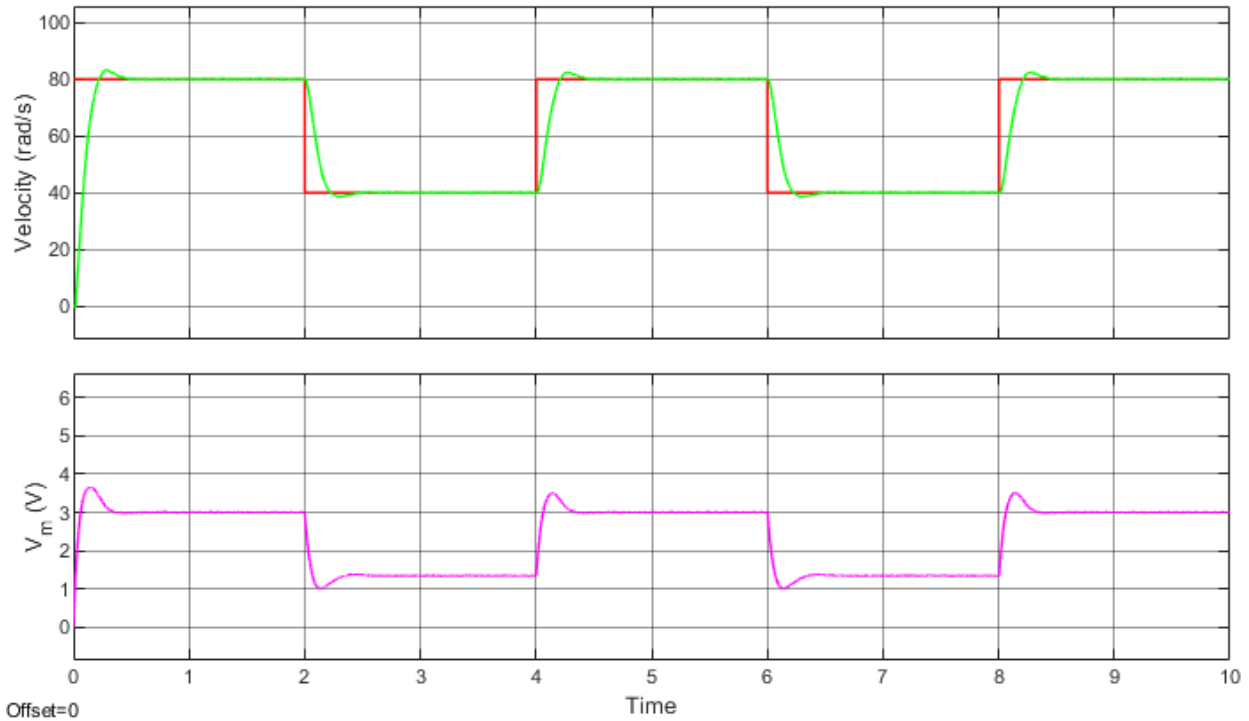
$$U(s) = K_i * E(s) - K_p * \Omega_m(s)$$

$$u(t) = K_p * e(t) - K_p * \omega_m(t)$$

2. System Response with the I+P Controller and predefined Gain Values

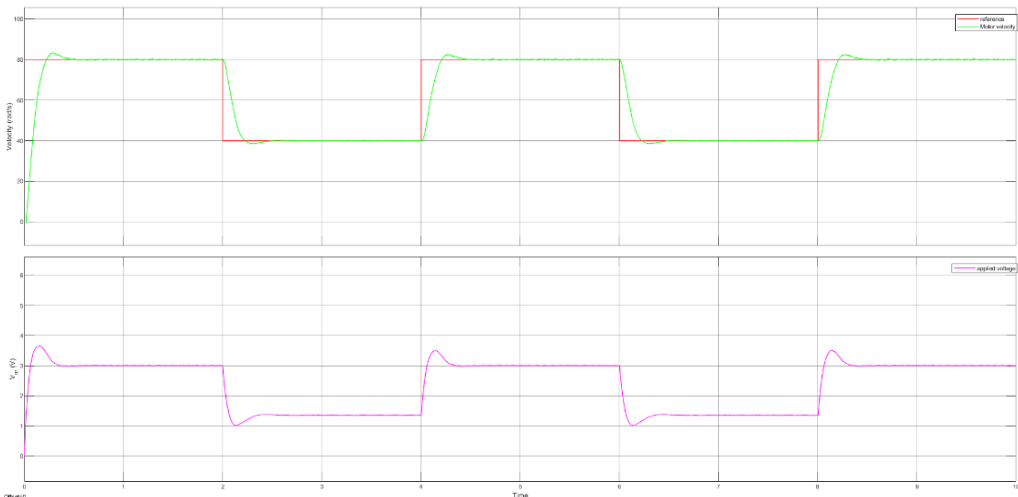
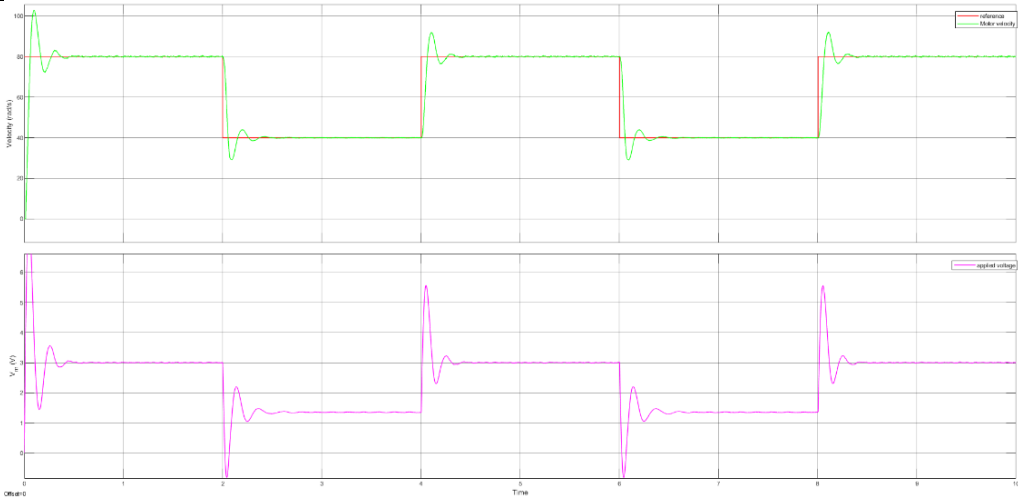


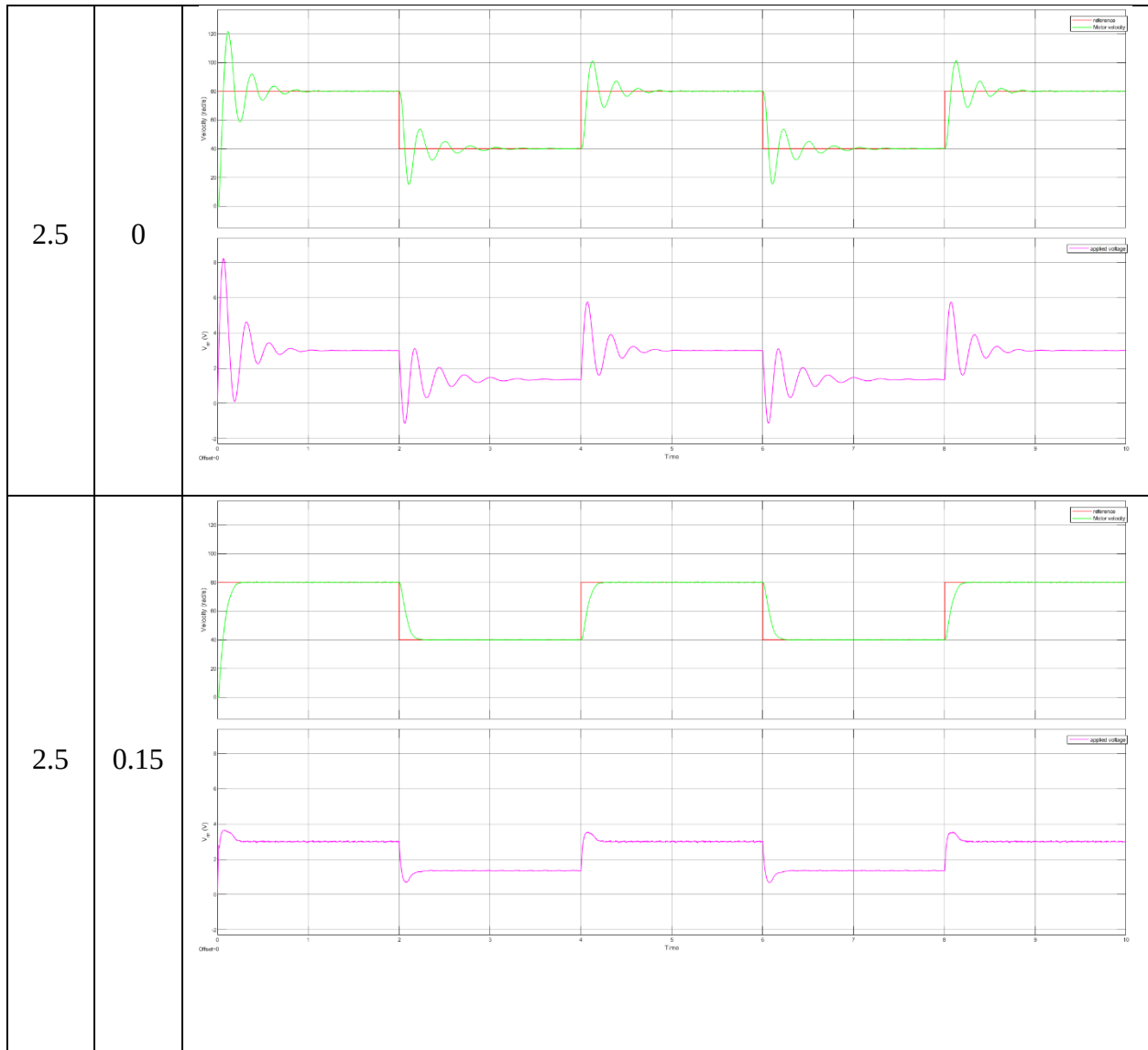
$K_i = 2.5 \text{ [V/(rad-s)]}$; $K_p=0.05\text{[V-s/rad]}$



3. Qualitative Analysis of the Controller

Different types of responses of the system were obtained as the controller gains were varied. Note that, as the K_i value increases, so does the overshoot and the peak time decreases. However, as the value of K_p increases, the overshoot value will be diminished.

Ki	Kp	System Response
1	0.05	
4	0.05	



4. Mathematical procedures to obtain the Closed-Loop Transfer Function.

a. Closed-Loop Transfer Function

b. Expressions for K_i and K_p in terms of the System Parameters

5. Matlab Script to compute Controller Gains

⑨ Find the K_p and K_v values

$$MP_d = 2.5\% ; t_{pd} = 0.15 \text{ sec} ;$$

$$\omega_{d1} = \omega_n \sqrt{1 - \zeta_d^2} ; t_p = \frac{\pi}{\omega_{d1}}$$

$$\zeta_d = \sqrt{\frac{\ln^2(MP_d)}{\ln^2(MP_d) + \pi^2}} \quad \text{eq. 1}$$

$$\omega_{d1} = \frac{\pi}{t_{pd}} \quad \text{eq. 2}$$

Solve eq. 1 and eq. 2 with $MP_d = 2.5\% ; t_p = 0.15 \text{ sec}$

$$\zeta_d = 0.7613 \quad \omega_{n1} = 32.3023$$

$$K_i = \frac{\omega_n^2 J_{eq} R_m}{K_t} ; K_p = \frac{2 \zeta \omega_n J_{eq} R_m}{K_t} - K_m - M$$

• Matlab 

$$K_i = 3.4032$$

$$K_p = 0.1202$$

```

%%System parameters
rm = 7.5;
kt = 0.0422;
km = 0.0422;
jm = 1.4e-6;
lm = 1.15e-3;
mh = 0.0106;
rh = 0.0111;
md = 0.053;
rd = 0.0248;
u = -2e-3; %Qube servo 3, SN: 56856
jh = 0.5 * mh * rh^2;
jd = 0.5 * md * rd^2;
jeq = jm + jh + jd;

%Open-Loop Transfer Fuction
num_op = kt/(jeq*rm);
den_op = [1 (km + u)*(kt/(jeq*rm)) 0];
G_op = tf(num_op,den_op);

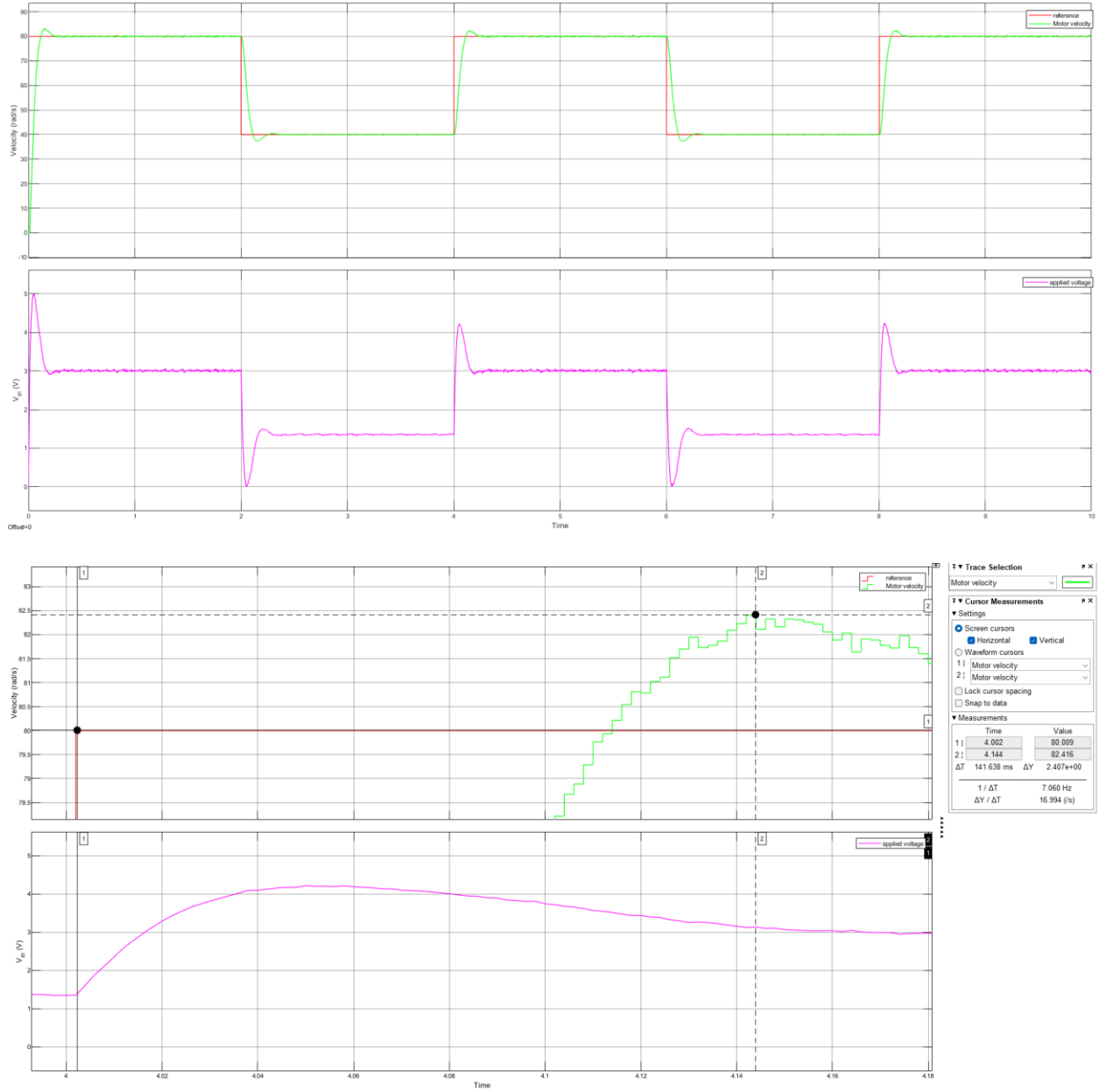
%%Desired response
tp_d = 0.15;
MP_d = 2.5/100;

zeta_d = sqrt(( log(MP_d) )^2 / (( log(MP_d) )^2 + pi^2));
wd_d = pi /tp_d;
wn_d = wd_d / sqrt(1-zeta_d^2);
den_d = [1 2*wn_d*zeta_d wn_d^2]; %Ch. equation
sigma_d = zeta_d*wn_d;

%Controller gains
%D.9
ki=wn_d^2*jeq*rm/kt;
kp=(2*zeta_d*wn_d*jeq*rm/kt)-km-u;

```

6. System Response with the I+P Controller with obtained Gain values



```
%Obtained data
%Original controller
A_ob=2.407;
B_ob=4e1;
MP_ob=A_ob/B_ob;%The controller by itself does not meet the MP requirement.
tp_ob=0.141613;
```

$$\%MP_{ob} = 6.02\%$$

$$t_{p_{ob}} = 0.142[\text{sec}]$$

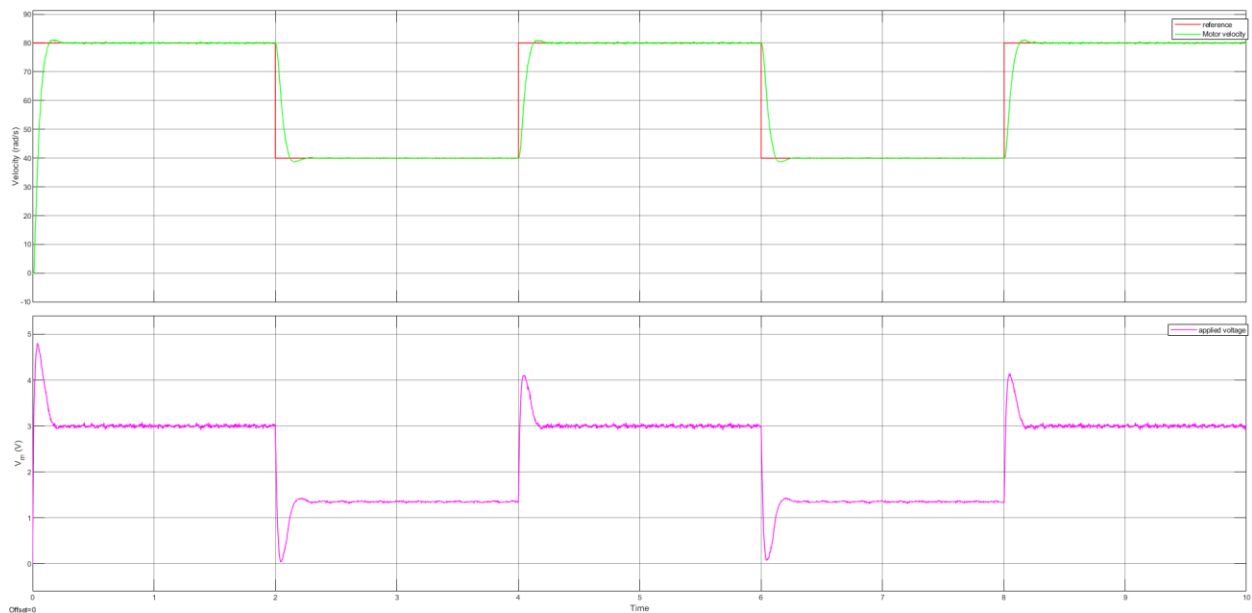
It can be seen in the previous figures that the I+P controller significantly improves the system response. However, the required overshoot is not satisfied.

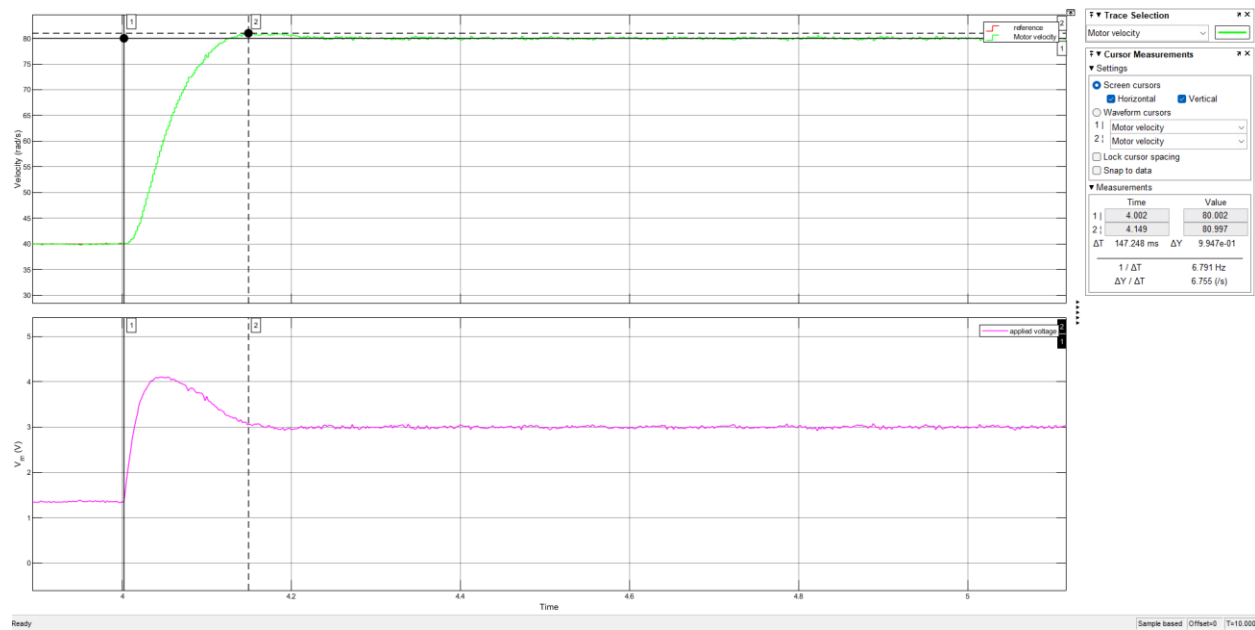
7. Tuning Procedure of the Controller Gains

The tuning procedure of the control gains K_i and K_p is the following:

1. Compute the Controller Gains directly from the system parameters.
2. Adjust K_p until achieve the desired overshoot ($k_p=0.15$).
3. Adjust K_i to obtain the desirable peak time ($k_i=4$).
4. Finally, adjust again K_p to achieve the desired overshoot ($k_p=0.16$).

The described procedure known as trial and error, let us achieve the desired performance indexes in the practical field.





```
%Modified controller for Kp and Ki
%Kp = 0.16
%Ki=4
A_mod=9.947e-1;
B_mod=4e1;
MP_mod=A_mod/B_mod;
tp_mod=0.147248;
```

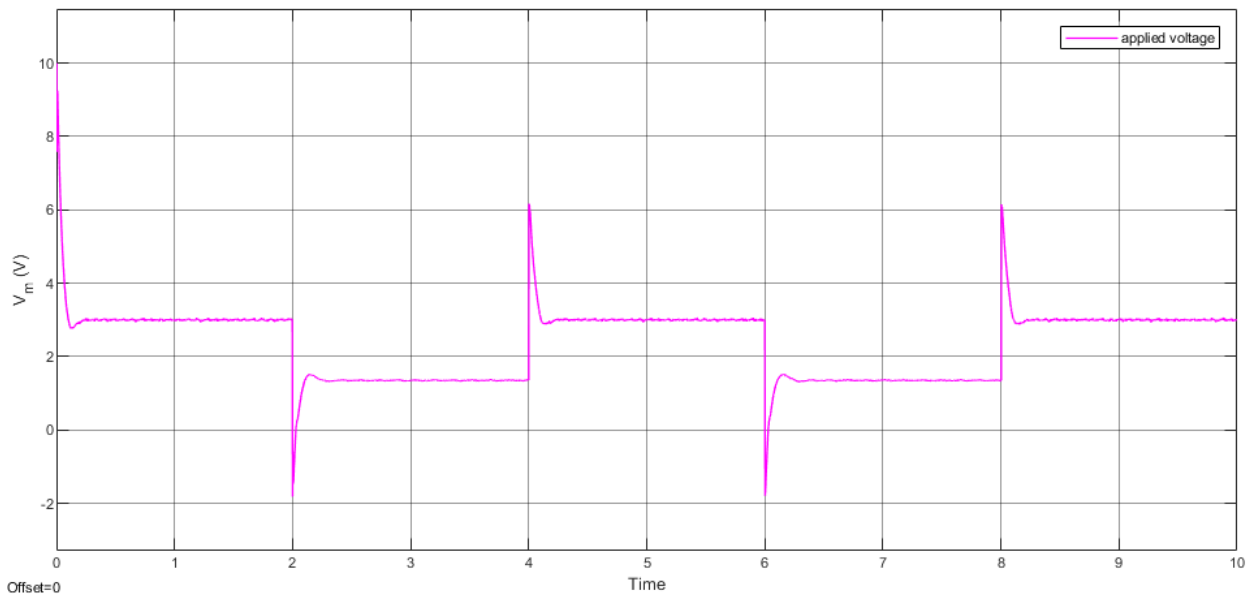
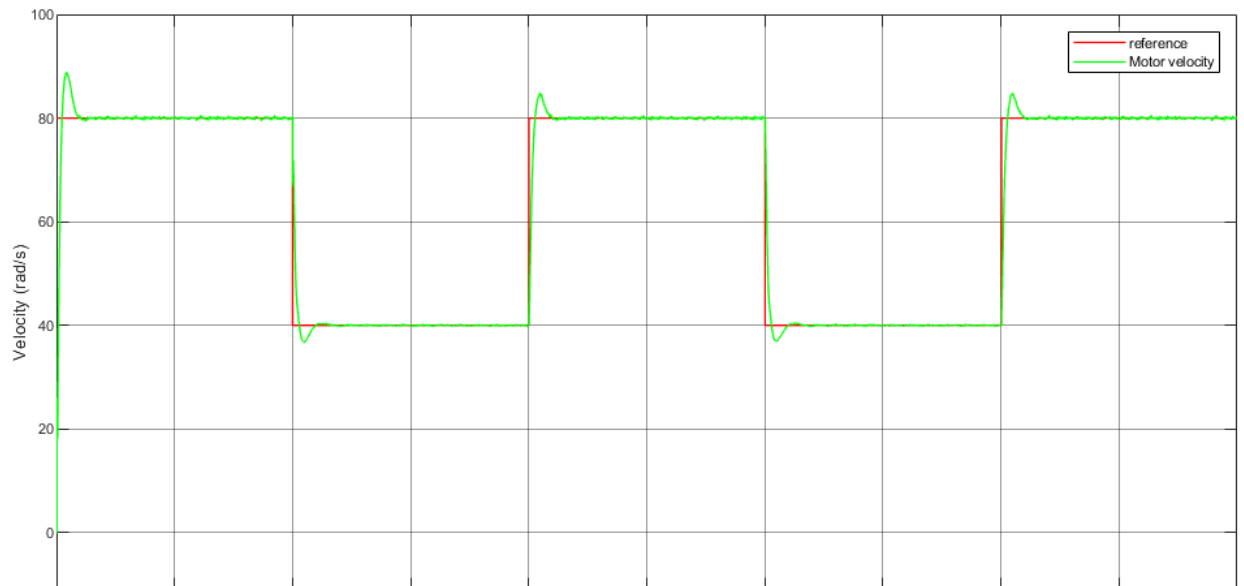
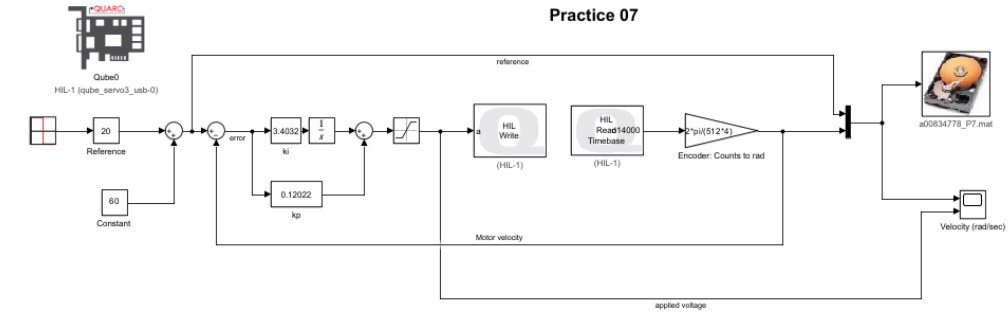
$$\%MP_{mod} = 2.49\%$$

$$t_{p_{mod}} = 0.147[sec]$$

8. PI Controller Analysis

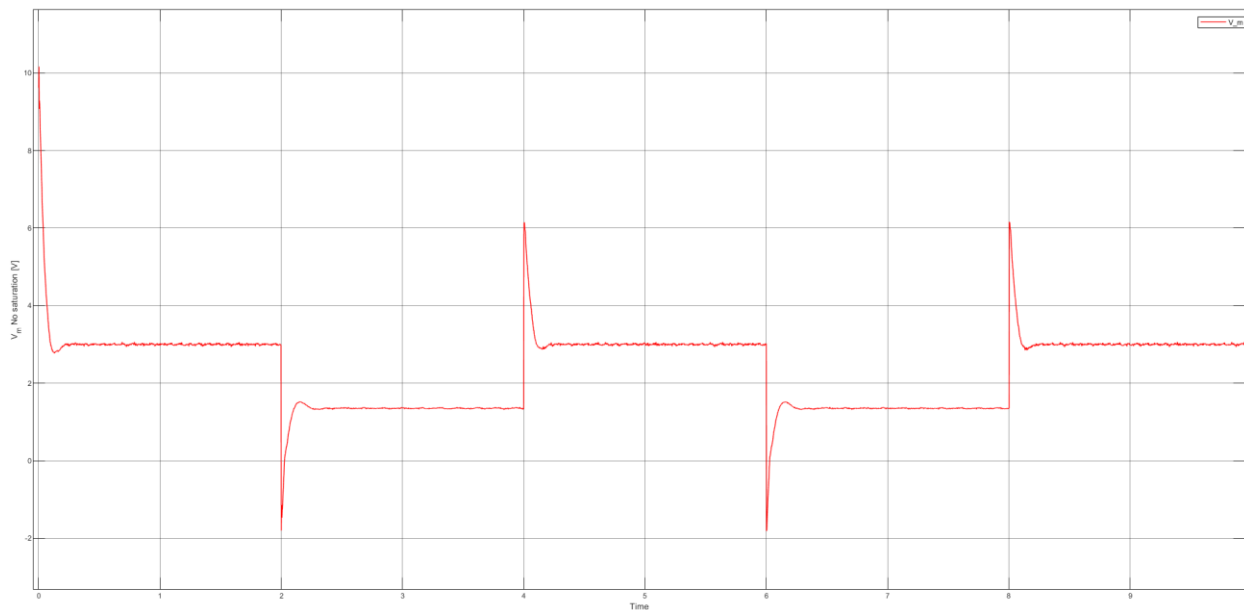
The main reason why the I+P controller is used instead of a PI controller is because of the setpoint kick phenomenon, where the manipulated variable achieves great values, resulting in reducing the time of life of the actuators in the practical field.

Practice 07



Offset=0

As we can see in the following figure, the manipulated achieves its maximum values of operation in short periods of time. Continuous operation under this conditions may cause damage in the actuators of the system.



9. Conclusions

In this practice, the integral plus proportional (I+P) control strategy was effectively applied to motor velocity control. Key outcomes include:

1. Performance Improvement: The I+P controller with tuned gains demonstrated reduced overshoot and peak time compared to the predefined values, though the peak time did not meet the initial expectations.
2. Tuning Insights: Adjusting K_i and K_p proved crucial, with higher K_p reducing overshoot and varying K_i affecting peak time. The trial-and-error tuning process improved performance indexes.

3. Comparison with PI Controllers: The I+P controller avoided issues like setpoint kick phenomenon that are common with PI controllers, making it more suitable for this application. And saving energy for the operation of the actuators and increasing their lifetime.

Overall, the I+P controller effectively managed motor velocity with room for further optimization through precise tuning, making it suitable for different practical purposes in comparison to a classical PI controller.